

UCREL overview: Natural Language Processing and Corpus Linguistics

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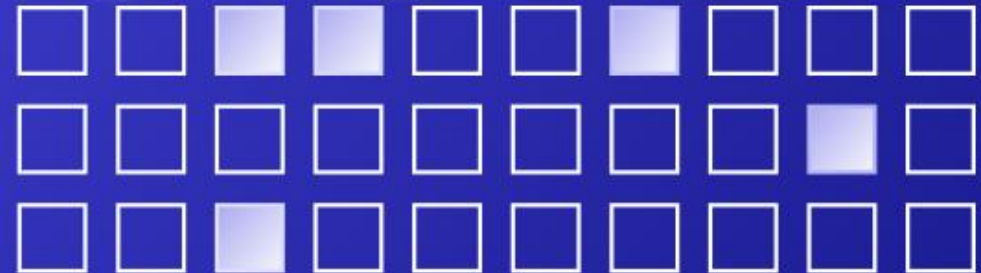
"Do you just count words?"

- "it's a little more complicated than that" 😊
- Many further considerations ...
 - what's actually a word?
 - MWEs
 - tokenisation
 - lemmatisation
 - spelling variation
 - meaning in context

UCREL Research Centre

- Incorporates postgraduate students, researchers, staff in *School of Computing and Communications* and *Linguistics and English Language Department* and many others
- Major themes: Corpus Linguistics and Natural Language Processing
- Applications in Social Sciences, Humanities, Health & Medicine and other disciplines where large quantities of text can be studied
- Key outputs and methodological contributions: compiled and linguistically annotated corpora, methods and software for corpus annotation, analysis and retrieval
- <https://ucrel.lancs.ac.uk/>

UCREL



Related centres and institutes

- ESRC Centre for Corpus Approaches to Social Science (CASS)
 - <http://cass.lancs.ac.uk/>
 - BNC2014, Climate Change discourse, Hate Speech, Language Teaching, Health Communication
- Lancaster Digital Humanities Centre
 - <http://wp.lancs.ac.uk/dighum/>
 - Spatial Humanities
- Data Science and AI Institute (DSAIL)
 - <https://www.lancaster.ac.uk/data-science-and-artificial-intelligence/>
- Security Centre
 - <http://www.lancaster.ac.uk/security-lancaster/>
 - Native Language Influence Detection (NLID), Language of extremism and counter-extremism, Author profiling, Online Child Protection
- FACTOR
 - Forensic linguistics, Cybersecurity and Technology Research
 - <https://wp.lancs.ac.uk/factor/>

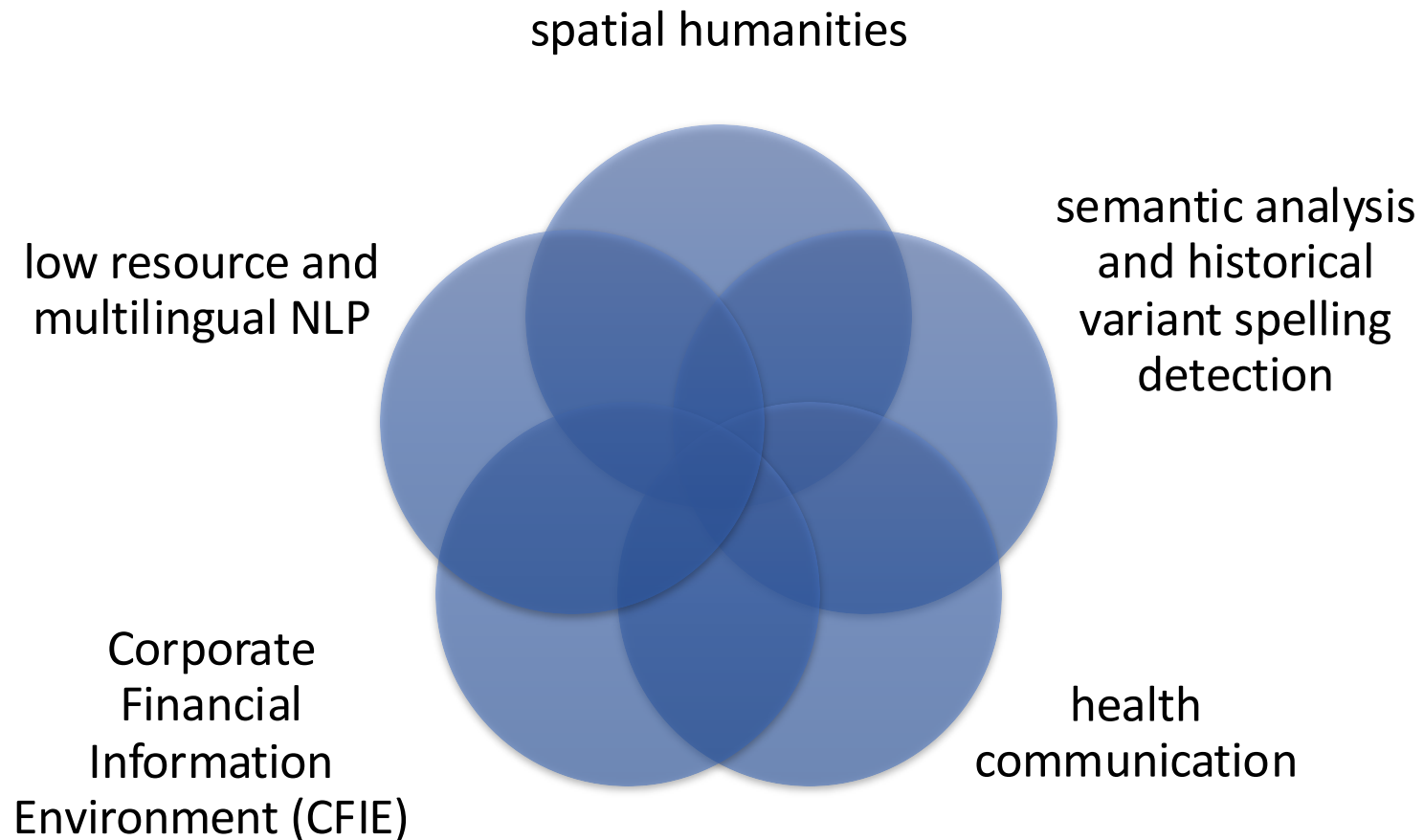


CLARIN ERIC

<https://www.clarin.eu/>

- CLARIN is a digital infrastructure offering data, tools and services to support research based on language resources
 - Virtual Language Observatory <https://vlo.clarin.eu/>
 - Switchboard <https://switchboard.clarin.eu/>
 - Federated content search <https://contentsearch.clarin.eu/>
- Many CLARIN resources and tools, including some of the core ones, are accessible to everyone, even outside of Europe. There is also the possibility to request a CLARIN identification if one's host institution is not federated
- ParlaMint project (<https://www.clarin.eu/parlamint>): 29 comparable corpora from 2015-23 parliamentary debates from around Europe

Five core areas ...

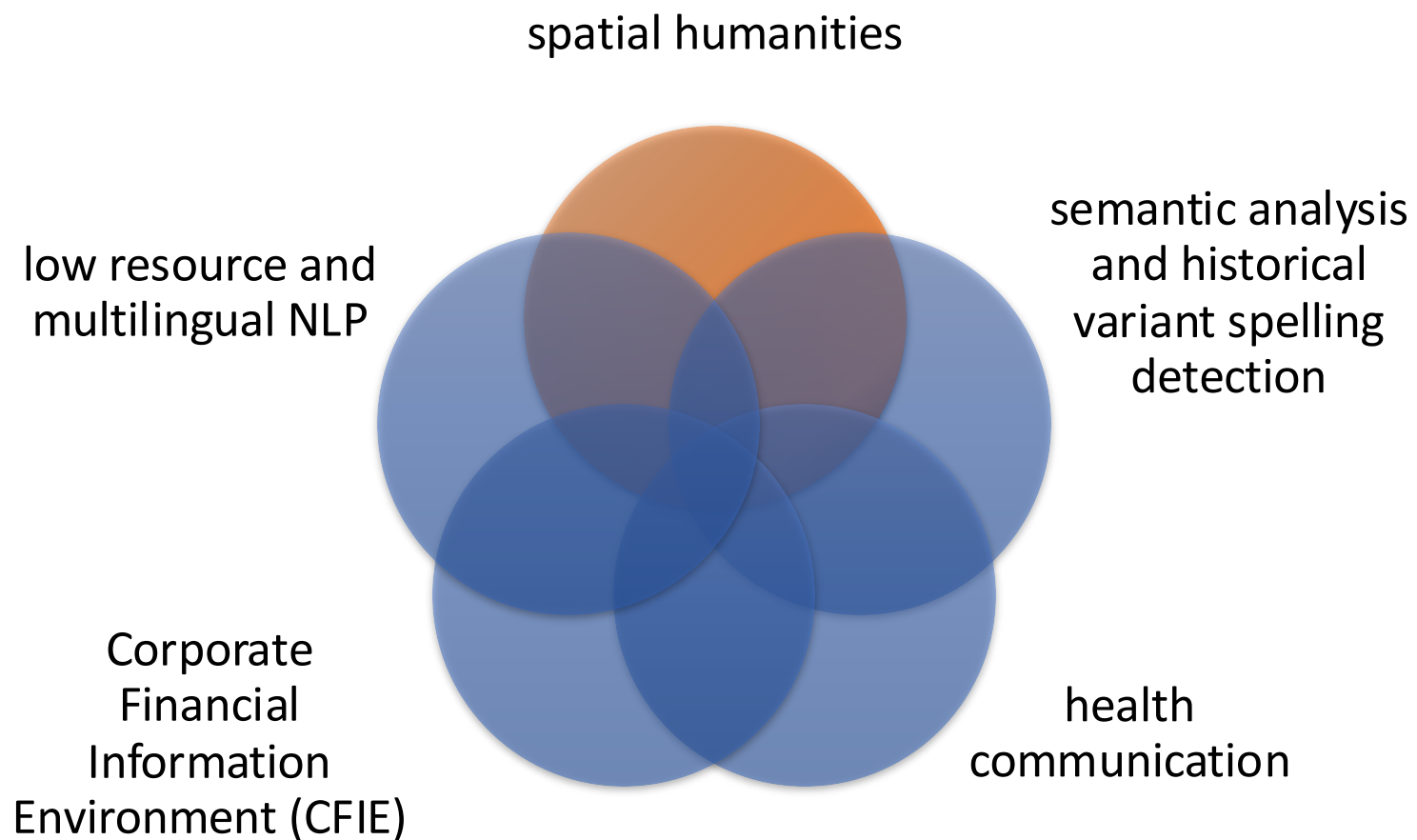


Additional or new areas ...

- Machine Translation, translation memory systems, anaphora resolution, computer-aided generation of multiple-choice tests
 - Ruslan Mitkov joined Lancaster in summer 2023
 - <https://wp.lancs.ac.uk/mitkov/>
- Security and Protection Science (SPS) initiative (phase 2; summer 2024)
 - <https://www.lancaster.ac.uk/scc/about-us/people/hansi-hettiarachchi>
 - <https://www.lancaster.ac.uk/scc/about-us/people/tharindu-ranasinghe>
- Software Engineering and NLP
 - Saad Ezzini and Jesse Phillips
- Efficient and low carbon NLP
- Sentiment and emotion analysis

Our Methodologies

- Natural Language Processing
 - Annotation: Part-of-speech tagging, Semantic Tagging, Word sense disambiguation
 - Variant spelling detection
 - Sentiment analysis
 - Text reuse
 - Named entity recognition
 - Language profiling
 - Unstructured text extraction
 - Dependency parsing
 - Information fusion
 - Network analysis
 - Topic modelling
 - Vector-based and NN approaches
 - Distributed / parallel processing
 - Machine Learning
 - GIS
- Corpus Linguistics
 - Frequency lists
 - Concordances
 - Keywords
 - N-grams / Clusters
 - Collocations

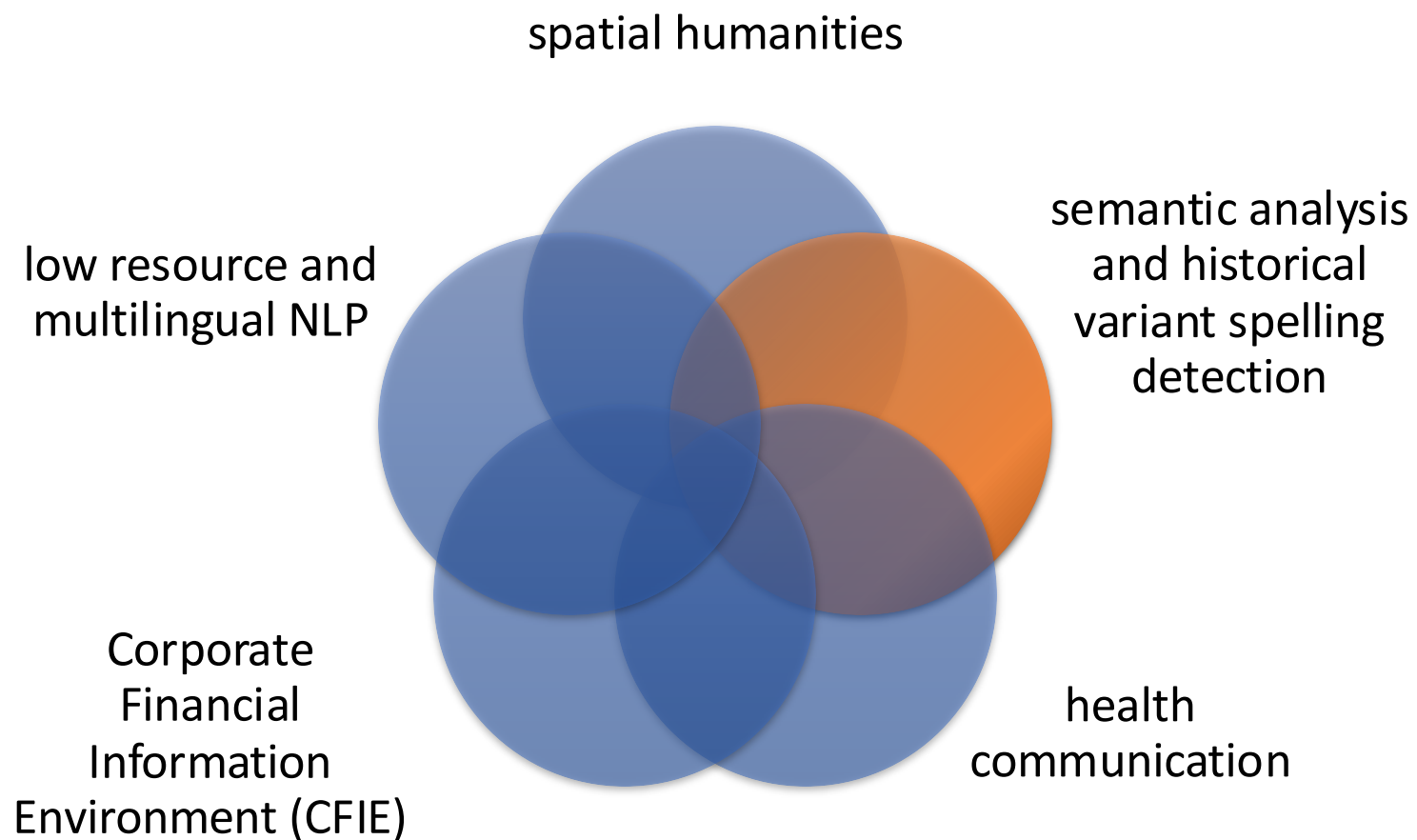


Spatial Narratives Project

<https://spacetime narratives.github.io/>



- Lancaster University: Ian Gregory, Paul Rayson, Ignatius Ezeani
- University of Leeds: Tony Cohn, John Stell, Erum Haris
- University of Bristol: Tim Cole
- University of Manchester: Jo Taylor
- Stanford University: Zephyr Frank, Erik Steiner
- Indiana University Purdue University, Indianapolis: David Bodenhamer, Neil Devadasan
- Datasets: Lake District travel writing, Holocaust survivor interviews
- The project is funded from 2022 to 2026 by ESRC (ES/W003473/1), and US National Science Foundation (NSF#2225179)



Example English semantic analysis

- *Last T1.1.1 year T1.1.1* was *A3+* the *Z5* UK *Z2* 's *Z5* second *N4* warmest *O4.6+++* on *A11.2+ record A11.2+*, *PUNC according Z5 to Z5* provisional *T1.3-* data *X2.2* from *Z5* the *Z5* Met *S3.1* Office *I2.1/H1c*. *PUNC* This *Z8 puts X2.2-* it *Z8* just *A14 behind X2.2-* 2022 *N1*, *PUNC* which *Z8* recorded *Q1.2* an *Z5* average *A6.2+* temperature *O4.6* of *Z5* only *A14* 0.06C *Z99* higher *N3.7++*. *PUNC*
- A3+ = being; A6.2 = comparing; A11.2 = importance; A14 = exclusivisers; H1 = architecture, buildings; I2.1 = business; N1 = numbers; N3.7 = measurement; N4 = linear order; O4.6 = temperature; Q1.2 = documents, writing; S3.1 = relationship; T1.1.1 = Time past; T1.3 = time period; X2.2 = knowledge; Z2 = geographical names; Z5 = grammatical bin; Z8 = pronouns etc; Z99 = unmatched

Though I **speake** with the tongues of men & of Angels, and **haue** not charity, I am become as sounding **brasse** or a tinkling cymbal. And though I **haue** the gift of **propheisie**, and **vnderstand** all mysteries and all knowledge: and though I **haue** all faith, so that I could **remoooue** **mountaines**, and **haue** no charitie, I am nothing...

(Authorised Version of the Bible, 1611)

I The external world

01 The world

- 01.01 The earth**
01.01.01 Region of the earth
01.01.02 Geodetic references
01.01.03 Direction
01.01.04 Land
01.01.05 Water
01.01.06 Named regions of earth
01.01.07 Structure of the earth
01.01.08 Minerals
01.01.09 Earth science
01.01.10 The universe
01.01.11 Atmosphere, weather
01.02 Life
01.02.01 Health and disease
01.02.02 Death
01.02.03 Biology
01.02.04 Plants
01.02.05 The body
01.02.06 Animals
01.02.07 People
01.02.08 Food and drink
01.02.09 Textiles

II The mental world

02 The mind

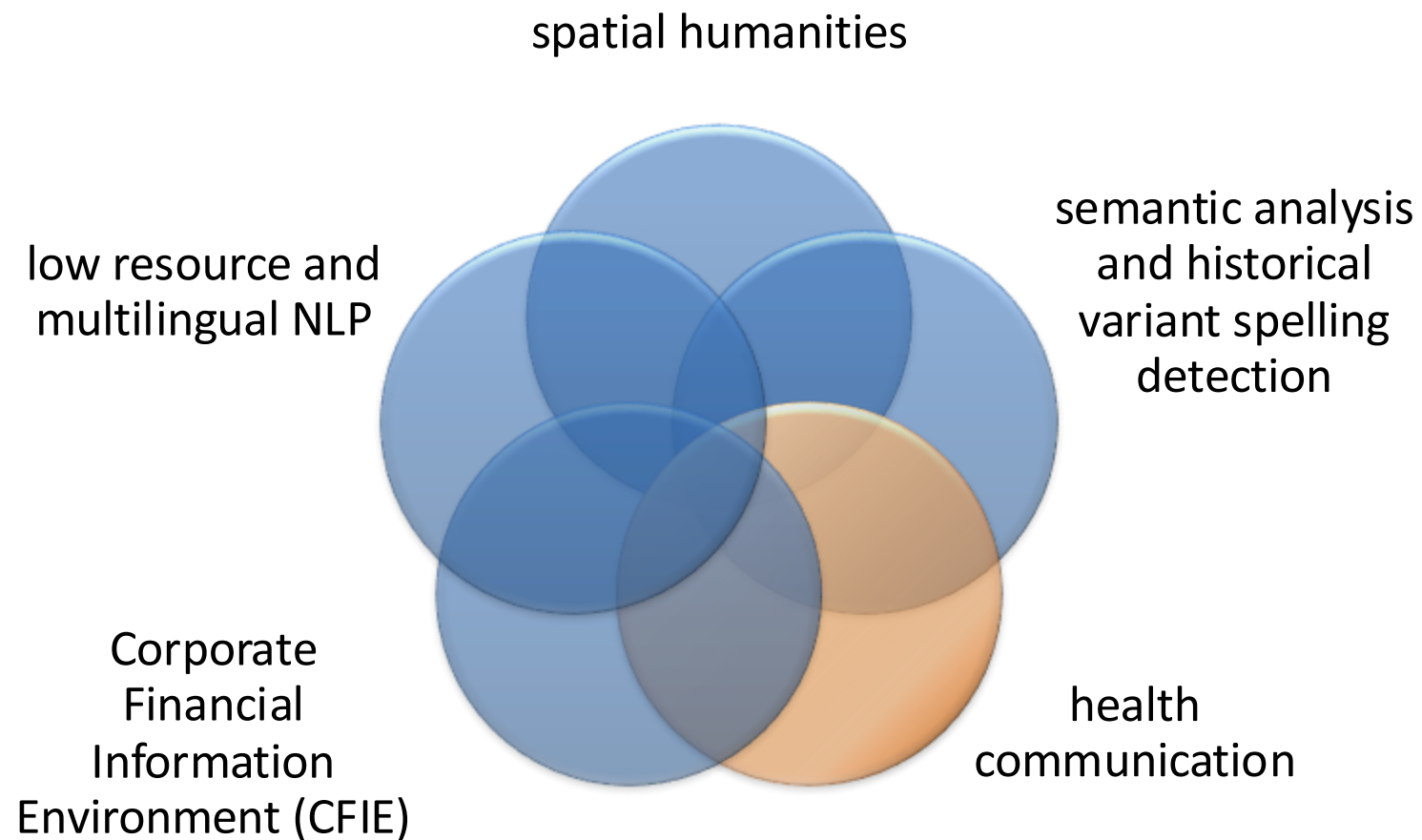
- 02.01 Mental capacity**
02.01.01 Spirituality
02.01.02 Intellect
02.01.03 Consciousness
02.01.04 Disposition/character
02.01.05 The psyche
02.01.06 Thought
02.01.07 Perception/cognition
02.01.08 Understanding
02.01.09 Lack of understanding
02.01.10 Intelligibility
02.01.11 Memory
02.01.12 Knowledge
02.01.13 Belief

III The social world

03 Society

- 03.01 Society/the community**
03.01.01 Kinship/relationship
03.01.02 Study of society
03.01.03 Society in relation to customs/values/beliefs
03.01.04 Social communication/relations
03.01.05 Social attitudes
03.01.06 Social class/rank
03.01.07 Dissension/discord
03.02 Inhabiting/dwelling
03.02.01 Inhabiting type of place
03.02.02 Inhabiting/dwelling temporality
03.02.03 Providing with dwelling place
03.02.04 Removing from dwelling place
03.02.05 Furnishing with inhabitants
03.02.06 Inhabitant/resident
03.02.07 Inhabited place
03.03 Armed hostility
03.03.01 War
03.03.02 Armed encounter
03.03.03 Victories and

A general and abstract terms	B the body and the individual	C arts and crafts	E emotion
F food and farming	G government and public	H architecture, housing and the home	I money and commerce in industry
K entertainment, sports and games	L life and living things	M movement, location, travel and transport	N numbers and measurement
O substances, materials, objects and equipment	P education	Q language and communication	S social actions, states and processes
T Time	W world and environment	X psychological actions, states and processes	Y science and technology
Z names and grammar			



Metaphor, cancer and end of life care (MELC) (2012-14)

- Analysis of metaphorical language used to talk about cancer, dying and death: people 'fight' their cancer, 'win' or 'lose' their 'battle' against it, hope for a positive end to their cancer 'journey', and so on.
- 1.5M word corpus of interviews and online forum posts from patients, carers and healthcare professionals
- Methods: Manual analysis (MIP) and Wmatrix (Semantic analysis & concordancing)
- <http://wp.lancs.ac.uk/melc/>

G3 Warfare (e.g. *fight* as a verb, *battle*)

A1.1.1 General actions, making (e.g. *blast*, *confront*)

A1.1.2 Damaging and destroying (e.g. *destroy*, *shatter*)

E3– Violent/angry (e.g. *hit*, *attack*)

S8+ Helping (e.g. *defend*, *protect*)

S8– Hindering (e.g. *fight* as a noun)

X8+ Trying hard (e.g. *struggle*)

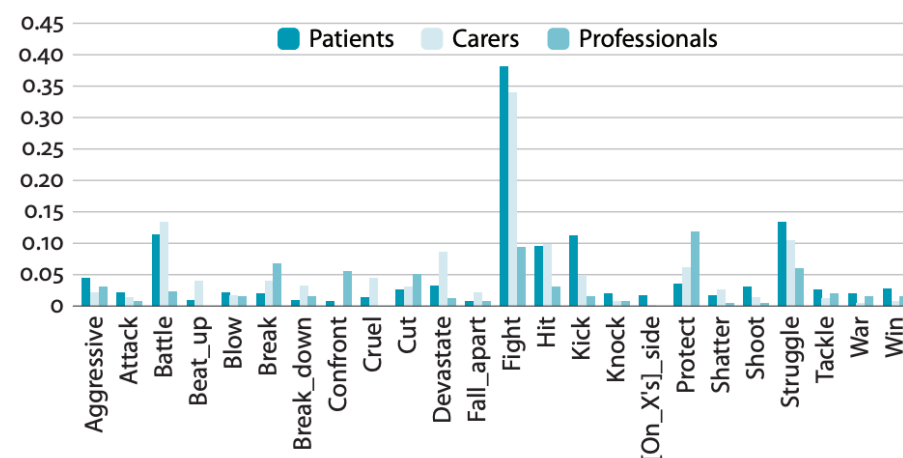


Figure 3. Relative use of most frequent Violence metaphors by each stakeholder group (per 1,000 tokens): Online forum posts

Software Architecture for Mental Health Self Management (SAMS)

- EPSRC working together project EP/K015796/1 (2013-16)
- Alistair Sutcliffe, Paul Rayson, Chris Bull and Pete Sawyer @ Lancaster
- Early diagnosis of dementia by monitoring interaction with a computer to monitor mental health
- circa 900K people in UK alone, only 44% receive a diagnosis
- combining data and text analysis
- <https://ucrel.lancs.ac.uk/sams/>



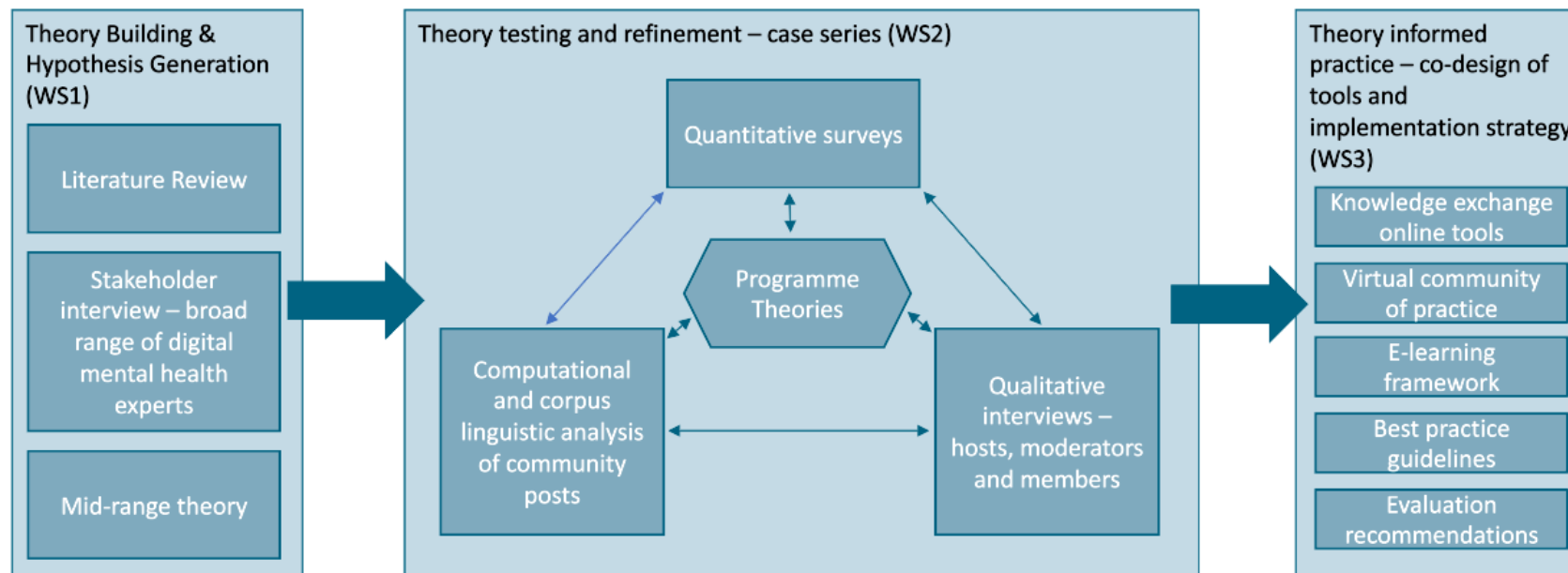
iPOF: improving peer online forums (2022-24)

<https://www.lancaster.ac.uk/ipof/>

How do online mental health forums work?

Why do some work better than others?

Why do some people find them helpful and others don't?



How can NLP and corpus linguistics be leveraged in realist evaluation?

Can data driven analysis inform development of realist programme theories?

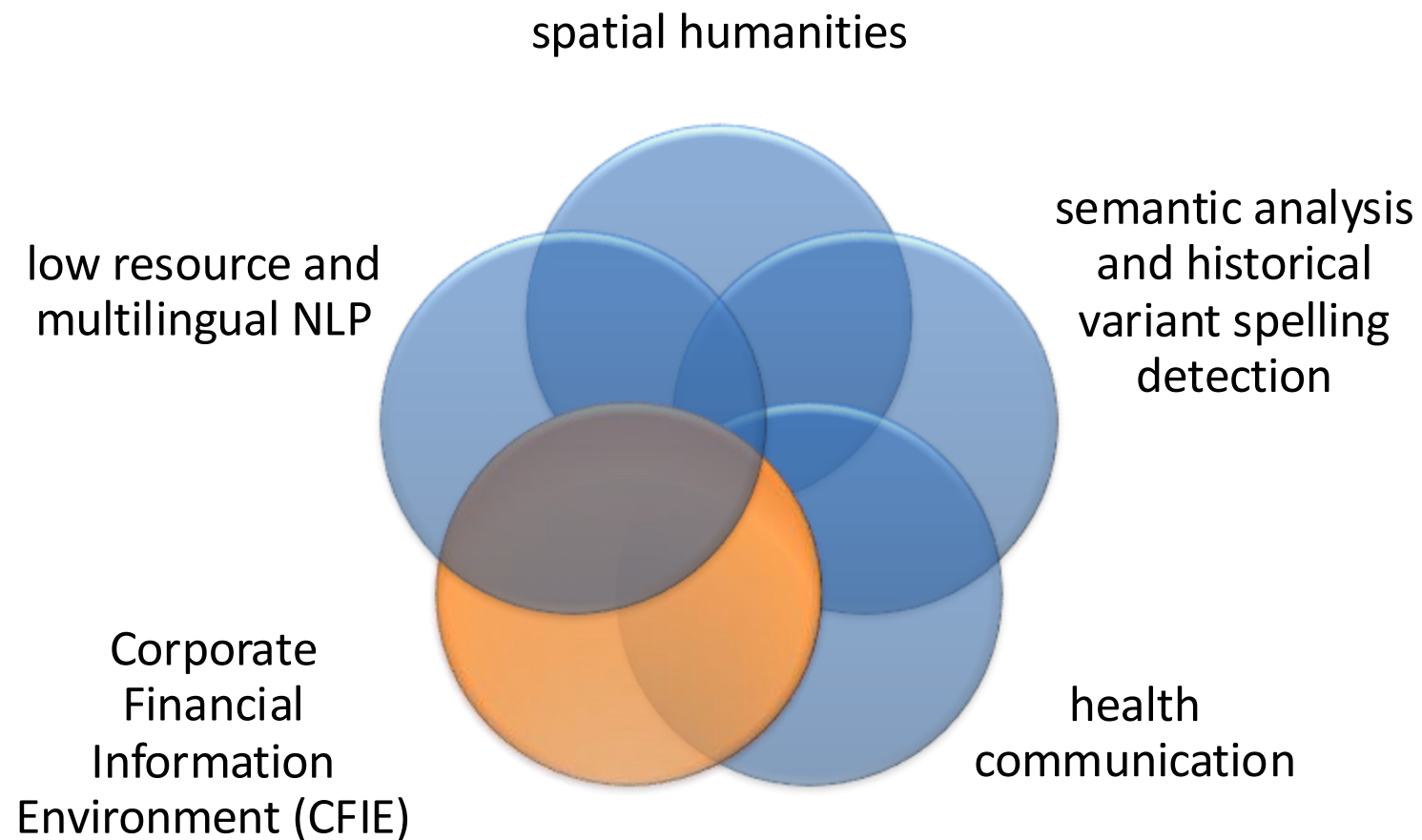
What characterises peer support forum data compared to other forums?

4D Picture (2022-27)

<https://4dpicture.eu/>



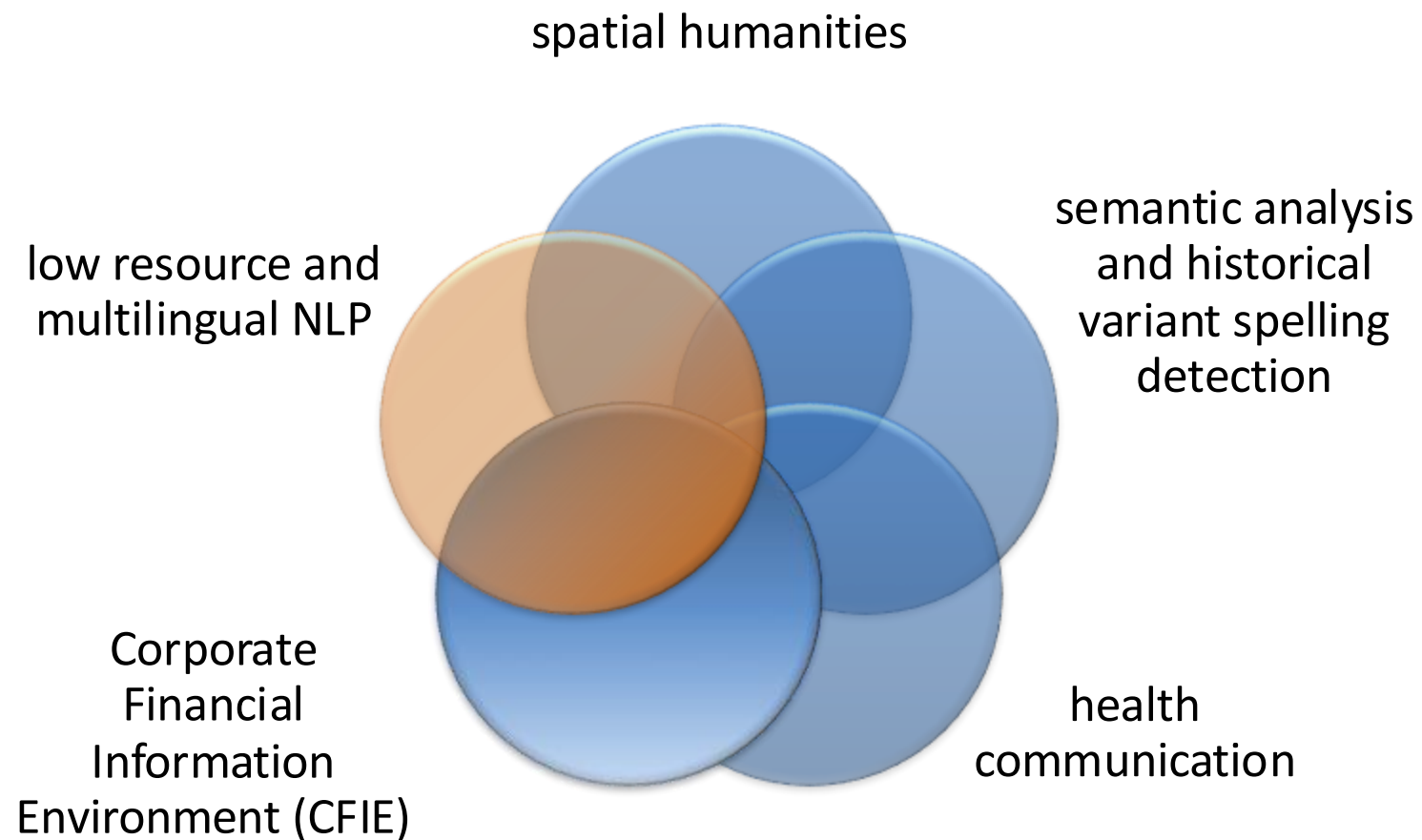
- The 4D PICTURE project aims to help cancer patients, their families, and healthcare providers better understand their options. It supports their treatment and care choices, at each stage of disease, by drawing on large amounts of evidence from different types of European data. The project involves experts from many different specialist areas who are based in nine European countries. The overall aim is to improve the cancer patient journey and ensure personal preferences are respected.
- Data collection and cleaning of corpora; NLP & ML prototypes; novel entity linking (citizen's vocabulary to professional discourse); aspect-based sentiment analysis; new version of Metaphor Menu



Corporate Financial Information Environment (CFIE) projects x 3



- ESRC, ICAEW, FRC, FCA funded three projects
 - Martin Walker, Manchester Business School
 - Steven Young, Lancaster University Management School
 - Paul Rayson, Lancaster University School of Computing & Communications
 - Mahmoud El-Haj, Lancaster University School of Computing & Communications
 - Vasiliki Athanasakou, London School of Economics
 - Thomas Schleicher, Manchester Business School
-
- Project seeks to analyse UK financial narratives, their association with financial statement information, and their informativeness for investors
-
- Automated, large sample analysis of UK annual report narratives represents a cornerstone of the project
 - Develop software for general use by academics
 - <https://ucrel.lancs.ac.uk/cfie/>





Lancaster
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Corpws Cenedlaethol Cymraeg Cyfoes
National Corpus of Contemporary Welsh

- Creation of a major language resource for Welsh speakers, Welsh learners, Welsh language researchers, and indeed anyone who is interested in the Welsh language. Multiple language samples will be gathered from real-life communication, and presented in a searchable online 'corpus', allowing users to explore Welsh as it is actually used.
- £1.8M funding (2016-19)
- <http://www.corcenc.org/>



2026 events we're (co-)organising

- The 2nd Workshop on NLP for Languages Using Arabic Script (AbjadNLP 2026) @ EACL2026
- Language Models for Low-Resource Languages (LoResLM2026) @ EACL2026
- The 7th Financial Narrative Processing Workshop (FNP 2026) @ LREC2026
- The Second Workshop on Holocaust Testimonies as Language Resources (HTRes-2026) @ LREC2026
- The 2nd International Workshop on Nakba Narratives as Language Resources (Nakba-NLP 2026) @ LREC2026
- Second International Conference on Natural Language Processing and Artificial Intelligence for Cyber Security (NLPAICS2026)
- International Conference New Trends in Translation and Interpreting Technology (NeTTIT'2026)
- 2nd International Workshop on Low Carbon Computing (loco2026)
- International Conference LAnguage TEchnologies for Low-resource Languages (LaTeLL2026)